

MMI 809 — APPLICATIONS OF DATA AND AI FOR DEVELOPMENT

Module 5: Ethics, Ownership, and Licensing in Development Data and AI

MSc in Mathematical Innovation • Open University of Kenya

1 MODULE OVERVIEW

Behind every AI system for development lie questions of power: Who owns the data? Who profits from the model? Who is harmed when it errs? Who can challenge its decisions? This module examines the ethical, legal, and political dimensions of development data and AI — including the mathematics of error (false positives and false negatives), governance structures, and the ownership frameworks determining whose interests AI ultimately serves.

LEARNING OUTCOMES — By the end of this module, you will be able to:

1. Critically assess AI tools and their asymmetric real-world impacts on different groups.
2. Explain the mathematics of false positives and false negatives and their unequal impacts.
3. Identify transparency, contestability, and governance gaps in AI systems.
4. Articulate ethical concerns around data ownership, consent, and benefit-sharing.

2 FALSE POSITIVES, FALSE NEGATIVES, AND ASYMMETRIC HARM

2.1 1.1 The Confusion Matrix

Actually Positive	True Positive (TP)	False Negative (FN)
Actually Negative	False Positive (FP)	True Negative (TN)

Key metrics: Sensitivity = $TP/(TP + FN)$; Specificity = $TN/(TN + FP)$; Precision = $TP/(TP + FP)$; Accuracy = $(TP + TN)/N$ (misleading for unbalanced data).

2.2 1.2 Asymmetric Harm by Application

Crop disease detection	Unnecessary treatment; wasted inputs.	Diseased crop untreated; harvest loss; food insecurity.
Credit scoring	Creditworthy person denied loan.	Uncreditworthy person given loan; over-indebtedness.
TB screening	Healthy person treated; side effects, stigma.	TB patient missed; continued transmission; mortality.

THE THRESHOLD QUESTION

Who decides where to set the decision threshold? This is not a technical question — it is an ethical and political one. Setting the threshold determines which error is more common and who bears it. Communities affected by the system should have voice in this decision.

3 DATA OWNERSHIP, ETHICS, AND GOVERNANCE

3.1 2.1 Data Ownership in Development Contexts

Data from LMICs creates significant value for AI developers and researchers. But the communities who generated the data rarely share in that value or control its use:

- Data is collected from communities, stored by organisations, and used by researchers — without benefit-sharing.
- Individuals rarely can revoke consent, see how their data is used, or challenge decisions made from it.

3.2 2.2 Algorithmic Accountability Principles

- **Transparency:** Existence and function of AI should be publicly known. Affected individuals should know when AI is used.
- **Explainability:** For high-stakes decisions, affected individuals should receive an explanation they can understand.
- **Contestability:** A mechanism must allow individuals to challenge AI-informed decisions.
- **Liability:** Clear accountability for AI harm must rest with identifiable human actors.

3.3 2.3 LLMs and Chatbots in Development: Specific Concerns

LLMs are increasingly marketed as development tools. Critical concerns:

- **Commercial bias:** Trained on English/high-income internet text; knowledge of LMIC contexts is often shallow or inaccurate.

- **Hallucination:** Generating plausible but incorrect information is particularly dangerous for health, agriculture, or legal guidance.
- **Accessibility:** Require good connectivity and English proficiency — precisely the barriers that most exclude target populations.
- **Data extraction:** Community queries may feed back to commercial developers without benefit-sharing.

4 ASSIGNMENT

MODULE 5 ASSIGNMENT: RESPONSIBLE INNOVATION REFLECTION (300–500 WORDS)

For your project: (1) Error analysis — what are the FP/FN scenarios and their harms? Who bears each error type? Where do you set the threshold and why? (2) Data ownership — who owns the data; who benefits; how will communities share in benefits? (3) Contestability — what mechanism allows affected individuals to challenge decisions? (4) Safeguards — what boundaries prevent harm from errors?

MODULE 5 SUMMARY

- False positives and false negatives have asymmetric real-world consequences, unequally distributed across demographic groups.
- The precision-recall trade-off is an ethical and political question, not a technical optimisation problem — communities should participate in threshold decisions.
- Data ownership is profoundly unequal: communities generate data; external actors extract value; individuals rarely have control or benefit.
- LLMs deployed as development tools carry commercial biases, hallucination risks, and accessibility barriers that require critical assessment.
- *The question is not whether AI is neutral. The question is whose values it embeds, whose errors it tolerates, and who has power to change it.*